

# **Script Assembler – Generation of Test Scripts for KeTop Devices**

## **DIPLOMA THESIS**

submitted for the

**Reife- und Diplomprüfung**

at the

**Department of Informatik**

Submitted by:

Filip Schauer  
Axel Csomány  
Adam Höllerl

Supervisor:

Adrian Kern

Project Partner:

Martin Wieser, KEBA Group AG

Leonding, April 7, 2026

I hereby declare that I have composed the presented paper independently and on my own, without any other resources than the ones indicated. All thoughts taken directly or indirectly from external sources are properly denoted as such.

This paper has neither been previously submitted to another authority nor has it been published yet.

The presented paper is identical to the electronically transmitted document.

Leonding, April 7, 2026

Filip Schauer , Axel Csomány & Adam Höllerl

# Abstract

Brief summary of our amazing work. In English. This is the only time we have to include a picture within the text. The picture should somehow represent your thesis. This is untypical for scientific work but required by the powers that are. Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.



# Zusammenfassung

Zusammenfassung unserer genialen Arbeit. Auf Deutsch. Das ist das einzige Mal, dass eine Grafik in den Textfluss eingebunden wird. Die gewählte Grafik soll irgendwie eure Arbeit repräsentieren. Das ist ungewöhnlich für eine wissenschaftliche Arbeit aber eine Anforderung der Obrigkeit. *Bitte auf keinen Fall mit der Zusammenfassung verwechseln, die den Abschluss der Arbeit bildet!* Suspendisse vel felis.

Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.



# Contents

|          |                                   |             |
|----------|-----------------------------------|-------------|
| <b>1</b> | <b>Introduction [X, F]</b>        | <b>1</b>    |
| 1.1      | Background [X] . . . . .          | 1           |
| 1.2      | Initial Problem [X] . . . . .     | 1           |
| 1.3      | Used Technologies [F] . . . . .   | 1           |
| <b>2</b> | <b>EBNF [A, F]</b>                | <b>2</b>    |
| 2.1      | Overview of EBNF [A] . . . . .    | 2           |
| 2.2      | Approaches to EBNF [A] . . . . .  | 2           |
| 2.3      | A Problem with EBNF [F] . . . . . | 2           |
| 2.4      | Our EBNF [F] . . . . .            | 2           |
| <b>3</b> | <b>Implementation [A, F, X]</b>   | <b>3</b>    |
| 3.1      | Preprocessor [A] . . . . .        | 3           |
| 3.2      | Parser [F, A] . . . . .           | 3           |
| 3.3      | Syntax Highlighting [A] . . . . . | 3           |
| 3.4      | Extensibility [F] . . . . .       | 3           |
| 3.5      | Database [X] . . . . .            | 3           |
|          | <b>Glossary</b>                   | <b>V</b>    |
|          | <b>References</b>                 | <b>VI</b>   |
|          | <b>List of Figures</b>            | <b>VII</b>  |
|          | <b>List of Tables</b>             | <b>VIII</b> |
|          | <b>List of Listings</b>           | <b>IX</b>   |
|          | <b>Appendix</b>                   | <b>X</b>    |

# **1 Introduction [X, F]**

## **1.1 Background [X]**

## **1.2 Initial Problem [X]**

## **1.3 Used Technologies [F]**

Just like the TRP project for which this project aims to provide an extension for, it was mainly written in the Python programming language. Aside from that, other tools have also been used or are otherwise important to be mentioned. The goal of this section is to give a broad overview of each of the technologies that we utilized or otherwise worked with when having written our project.

### **1.3.1 Python**

Python is a high-level, interpreted, dynamically typed object oriented programming language with a focus on allowing programmers to create applications, scripts, or connections between existing programs, quickly.[1]

## **2 EBNF [A, F]**

### **2.1 Overview of EBNF [A]**

### **2.2 Approaches to EBNF [A]**

### **2.3 A Problem with EBNF [F]**

### **2.4 Our EBNF [F]**

## **3 Implementation [A, F, X]**

### **3.1 Preprocessor [A]**

### **3.2 Parser [F, A]**

### **3.3 Syntax Highlighting [A]**

### **3.4 Extensibility [F]**

### **3.5 Database [X]**





# Glossary

**TRP** TestRail Procedure

1

# References

- [1] Python Software Foundation, “What is Python? Executive Summary,” retrieved 2026-02-21. [Online]. Available: <https://www.python.org/doc/essays/blurb/>

# List of Figures

# List of Tables

# List of Listings

# Appendix